

Leveraging Nonverbal Signals for Hands-Free Input in Digital Surveys

Chandan Kumar

chandan.kumar@iao.fraunhofer.de

Human-Technology Interaction

Fraunhofer Institute for Industrial Engineering IAO

Stuttgart, Germany

Apeksha Rahangdale

apeksharahangdale02@gmail.com

Fraunhofer Institute for Industrial Engineering IAO,

RWTH Aachen University

Germany

Eileen Wemmer

eileen.wemmer@iat.uni-stuttgart.de

Institute of Human Factors and Technology Management

University of Stuttgart

Stuttgart, Germany

Astrid Rosenthal-von der Pütten

arvdp@itec.rwth-aachen.de

Chair for Individual and Technology

RWTH Aachen University

Aachen, Germany

ABSTRACT

Collecting end-user feedback is crucial for assessing the acceptance and viability of any product or service. In the digital realm, Likert scale surveys are the most common method for gathering feedback, with end users explicitly selecting their responses using a mouse or touch input. Conversely, in social environments, nonverbal interactions are often used as an intuitive means of expressing agreement or disagreement. In this work, we explore the utilization of nonverbal interactions such as facial expressions and head gestures as a novel approach to elicit user responses in digital surveys. This presents a valuable hands-free alternative for obtaining public feedback in a non-intrusive manner, particularly enabling individuals with motor impairments to participate in surveys. We outline our framework and the evaluation study with 16 participants to assess the acceptance and feasibility of the proposed hands-free method for responding to Likert scale survey questions.

CCS CONCEPTS

• **Human-centered computing** → **Interaction techniques**; **Gestural input**; *User studies*; *User centered design*; **Accessibility technologies**.

KEYWORDS

multimodal interaction, hands-free interaction, likert scale input, facial expressions, head movements, digital survey

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1 INTRODUCTION

Over the past two decades, digital surveys have witnessed widespread adoption across product design, development, and research initiatives. Specifically, Likert scale-based questions, designed to collect structured responses indicating agreement or disagreement with specific statements, have become a prominent survey methodology [2]. Aligned with the principles of universal design and inclusiveness, there is a growing recognition of the need for digital surveys to cater to a diverse range of users, irrespective of their abilities and environmental limitations [8, 11]. However, the conventional methods of responding to multiple Likert scale questions in digital surveys often rely on explicit methods such as mouse or touch input. This poses a significant challenge for individuals with limited mobility, motor impairments, or those in situations where manual interaction may be impractical or unsafe. Examples of the latter include scenarios such as driving, multitasking, or interacting with shared public displays. Although voice-based input serves as an apparent alternative [13, 22], its usage is subject to user ability, recognition errors, and privacy concerns. This underscores the need for alternative, nonverbal, hands-free input methods to ensure accessibility and inclusivity.

In the domain of nonverbal hands-free interaction, various modalities have been investigated [8]. However, facial expressions and head gestures emerge as particularly natural means of conveying human agreement or disagreement, a concept extensively discussed in communication and social conversation literature [16, 17, 26]. Furthermore, recent years have witnessed notable advancements in face and head gesture technology, showcasing increased robustness and accuracy, as well as successful adoption across diverse domains, encompassing mouse replacement, gaming, and assistive technology [4, 9]. This highlights the potential of integrating face and head-gesture-based interactions into digital surveys. Despite this potential, existing research has not thoroughly explored the design and implementation of such interactions. Our objective is to address this gap by proposing face and head-gesture-based interactions as a survey interaction modalities and assessing their feasibility and acceptance.

In pursuit of this objective, this paper presents the development of an initial web-based prototype interface that incorporates facial

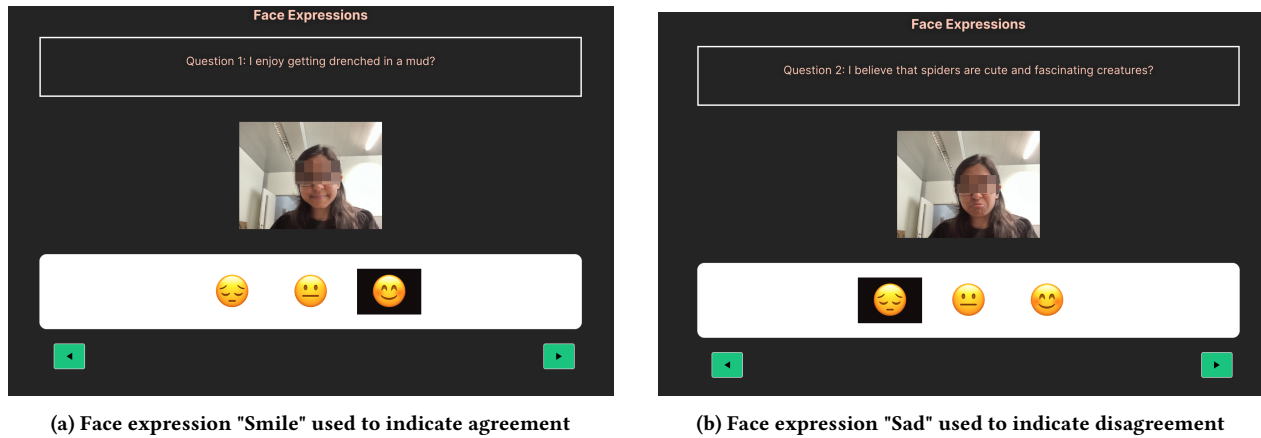


Figure 1: Interface Screenshots Capturing Facial Expression Input on a 3-Point Likert Scale

expressions and head gestures for inputting responses in 3-point and 5-point Likert scale questionnaires. To evaluate the viability and acceptance of these modalities among end users, we conducted a lab study involving 16 participants. The results signify the potential of these modalities as alternative input methods, marking a crucial milestone for our future research to develop and assess these modalities in more robust and real-world survey environments.

2 RELATED WORK

2.1 Hands-free interaction for Surveys

Hands-free interaction in digital surveys has gained considerable importance for fostering inclusivity, enhancing user experience, and optimizing data collection processes. Hands-free methods are important in ensuring accessibility for individuals with motor impairments or disabilities [8, 20]. This aligns with the principles of universal design, emphasizing the need for digital surveys to accommodate a diverse range of users [6, 19]. As a hands-free method, voice interaction has been predominant in obtaining survey responses on web and mobile platforms [13, 22, 23]. Still, it faces limitations such as dictation errors and privacy concerns. The exploration of nonverbal means, such as face, head, and body gestures, has been studied in face-to-face interviews [9, 14]. However, its applicability in filling digital surveys remains an area that requires further exploration.

2.2 Facial expressions and head gestures in communication

In social conversations, facial expressions and head gestures play a pivotal role in conveying nuanced communication beyond verbal language. Recent studies [16, 17] emphasize the role of facial expressions as powerful indicators of emotional states. Similarly, head gestures have been explored in the context of social communication, with studies [25] shedding light on the subtle cues provided by head movements during conversations. The combination of facial expressions and head gestures [4, 9] has also been identified as an influencing factor in the interpretation of emotions, intentions, and agreement levels in social interactions. Beyond head gestures, eye

movement and eye contact have been acknowledged as vital characteristics in evaluating user engagement [18]. However, it remains challenging to ascertain agreement or disagreement through eye contact.

2.3 Facial expressions and head gestures as input methods

The utilization of facial expressions and head gestures as input methods in human-computer interaction (HCI) has gained prominence due to their potential to enhance user experiences and interaction modalities. The work by Ekman and Friesen [10] laid the foundation for understanding facial expressions as conduits for conveying emotions, with subsequent research exploring their application in technology interfaces such as in assistive technology [21] and gaming [24]. Yan et al. [10] leveraged facial expression changes in response to unexpected interactions for smart speaker interactions. Similarly, the effectiveness of head gestures as an input method has been recognized in diverse applications, ranging from virtual reality environments [30] for answering yes or no questions, to interactions in gaming [27] and assistive technologies [28]. Multi-modal approaches that assess both face and head as input methods have also been explored, such as Chamin and Lyons' [29] design and evaluation of vision-based head and face tracking interfaces for assistive input means. In the multimodal game "Take Drunkard Home" [15] head movements control the movement of a drunken character, while facial expressions influence the behavior of other in-game characters, with positive expressions leading to smiling and negative expressions to frowning.

3 PROTOTYPE INTERFACE

Figure 1 depicts the prototype developed to facilitate user input within a Likert scale questionnaire environment. The central region of the interface prominently features the user's reflection captured by the webcam, offering immediate and intuitive feedback on their facial expressions or head gestures. This real-time visual representation serves to enhance the user's awareness and engagement during the survey process. In the top section, the interface displays the questions presented as part of the questionnaire survey. The bottom

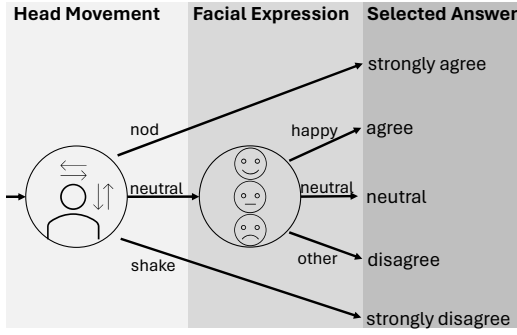


Figure 2: To select an answer for 5-point scales, head movements are used to input *strong (dis-)agreement*, while a still head corresponds to *agree, neutral or disagree*, depending on the facial expression.

area is dedicated to showcasing the agreement scale, presented as a 3 or 5-point Likert scale. To make the scale more relatable and user-friendly, smileys are employed as visual representations. The choice to use smileys is grounded in their relatability to human natural expressions, supported by prior research highlighting that the incorporation of emojis, including smileys, can have a positive impact on user engagement during communication and survey participation [1, 3, 5]. To ensure platform independence, the interface was created as a web application utilizing the React framework¹.

The process of capturing facial expressions involved utilizing the webcam and extracting a continuous stream through of emotion class likelihoods using face-api², which recognizes the six basic emotion categories as proposed by Ekman [7] plus a *neutral* category. Emotion classes were assigned to positive, neutral, and negative based on the Likert scale value they likely indicate. More specifically, *happy* was associated with agreement, *neutral* with neutral and all remaining emotion classes (*sad, angry, fearful, disgusted, surprised*) were linked to disagreement.

Detection of head movements, including *nod* for "agree," *neutral* for neutral, and *head shake* for "disagree," commenced with the extraction of face landmarks through MediaPipe³. By tracking the movements of the chin landmark alongside a landmark marking the left side of the face, calculations were performed to determine if the head engaged in a nodding, shaking, or neutral motion. Scaling values were adjusted based on the participant's distance from the screen, estimated from the distance between the landmarks for the top and bottom of the head. The magnitude of the movement was then compared to heuristically determined thresholds to identify the specific head movement executed. To respond to five-point Likert scale questions, both facial expressions and head movements was used. Nodding indicated strong agreement, shaking the head indicated strong disagreement, while facial expressions were employed to choose between agree, neutral, or disagree. This selection process is visualized in figure 2.

¹<https://react.dev/>

²<https://justadudewhohacks.github.io/face-api.js/docs/index.html>

³<https://github.com/google/mediapipe/wiki/MediaPipe-Face-Mesh>

4 EXPERIMENTAL EVALUATION

To evaluate the feasibility of leveraging head gestures and facial expressions in conjunction, we conducted a small-scale experimental study in a lab setting. For comparability, this entailed the participants answering the same questionnaire repeatedly on both three-point and five-point scales, using varying hands-free input mechanisms. For three-point scales, either head movements (*nod, shake, neutral*) or facial expressions (*joy, neutral, sadness*) were considered. To extend the information value of these modalities for five-point scales, we combined the inputs. Furthermore, to assess this novel input mechanism for Likert scales in comparison to a widely adopted option, participants also answered five-point questionnaires using voice input.

4.1 Participants

The study was conducted with a total of 16 participants between the ages of 22 and 27 (mean=24.63, SD=1.63). 11 participants were male, 5 were female. While most participants were students (13), some were employed full-time (2) or currently unemployed (1). We gathered prior experience in using face, head or voice for answering questionnaire questions on a five-point scale. In line with our expectations, both head movements (mean=1.75, SD=1.0) and facial expressions (mean=1.63, SD=0.96) were less commonly encountered by participants in the context of surveys in the past than voice input (mean=2.75, SD=1.69). This contrasts with how participants described their use of the same modalities to convey agreement or disagreement in social situations. When asked how often each modality was used in social situations, all three options showed more comparable scores. While the usage of head gestures was rated the least common with a mean of 3.5 (SD=1.21), facial expressions (mean=4.13, SD=1.31) were perceived to be employed slightly more often than voice (mean=4.0, SD=1.37). This difference highlights the potential of head movements and facial expressions as natural input mechanisms to convey (dis-)agreement.

4.2 Implementation and Apparatus

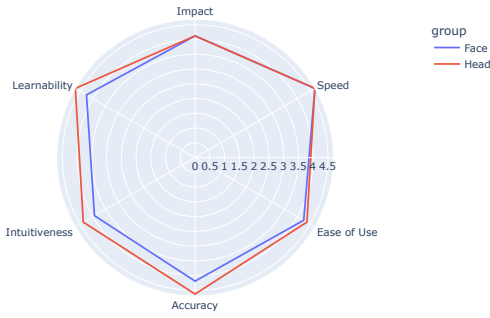
The study was conducted on an a laptop (3.20 GHz CPU, 8GB RAM, M1 chip) running iOS 13. The built-in webcam and microphone were used for capturing participants' head and face movements and their voice inputs respectively. The details on face and head input has been elaborated in Section 3. For all modalities, the input was controlled by pressing and releasing the space bar to start and stop evaluation respectively. For voice input, this started a voice recording which was then passed to a speech-to-text API⁴.

4.3 Procedure

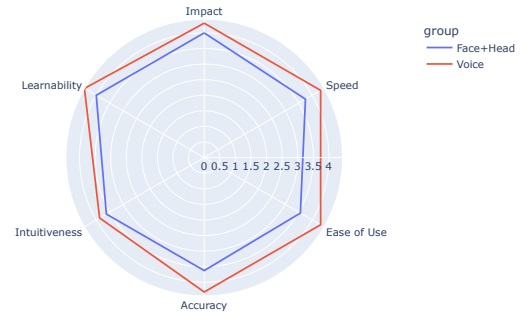
The study was conducted at the University of Stuttgart. Upon arrival, participants were greeted and provided detailed information about the experiment procedure, confidentiality, and payment. Participants were then asked to fill out a demographic survey which included their prior experience with the relevant input mechanisms in the context of surveys.

The main study consisted of the participants answering the same ten questions, split into two sessions of five questions each, in

⁴<https://www.npmjs.com/package/react-speech-recognition>



(a) Usability Scores for 3-point scale input mechanisms: Facial Expressions and Head Gestures



(b) Usability Scores for 5-point scale input mechanisms: Combined Face and Head input, and Voice

Figure 3: Mean usability scores for all modalities.

the same order each time. The experiment was conducted within-subject in two blocks: The first block focused on head movements and facial expressions as input mechanisms for three-point Likert scales. The second considered the bi-modal head and face input, as well as voice for answering five-point Likert scale questions. Both blocks started with the participants answering all ten questions through a conventional, mouse-based form. While the order in which the modalities were presented was randomized within blocks and each modality appeared first equally often, this assured that participants would not need additional time to consider the questions and answers for the first modality. For each modality there was a training session and two main sessions. After input with each modality was completed, participants gave feedback about the input mechanism they had just used through a questionnaire. During training, participants were given up to seven attempts to answer each question. In the main sessions maximum trials were limited to 3. After the completion of the main session for each modality, participants provided feedback on a total of six usability measures through a survey. In addition, the survey included the NASA-TLX [12] questionnaire. All feedback was collected on a five-point Likert scale. After the main study, participants were asked to feedback evaluating all four input mechanisms.

4.4 Results

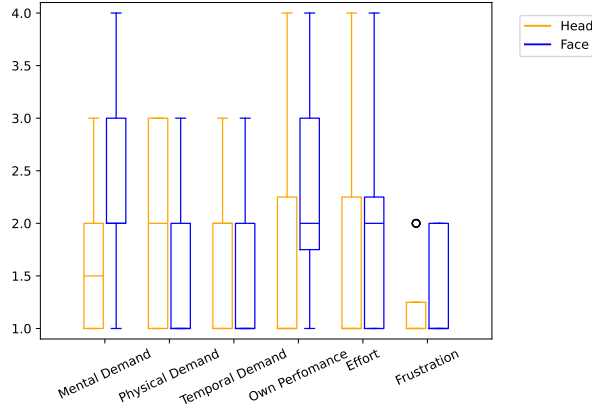
Usability. After participants filled out the questionnaire with each modality, they provided feedback on various usability measures on a five-point Likert scale ranging from 1 to 5. We report the mean usability scores in figures 3a and 3b for modalities used on three-point and five-point scales respectively. Overall, we receive positive user feedback across all modalities, with mean scores consistently above 3.5 for all measures and modalities, indicating that participants were able to effectively utilize the proposed interaction modalities for completing Likert scale surveys.

When responding to three-point Likert scale questions, mean scores for head movements slightly surpass those for facial expressions. For five-point scales, voice input garnered slightly better responses than the bi-modal face and head input. It is worth noting, however, that this trend persists despite participants having more

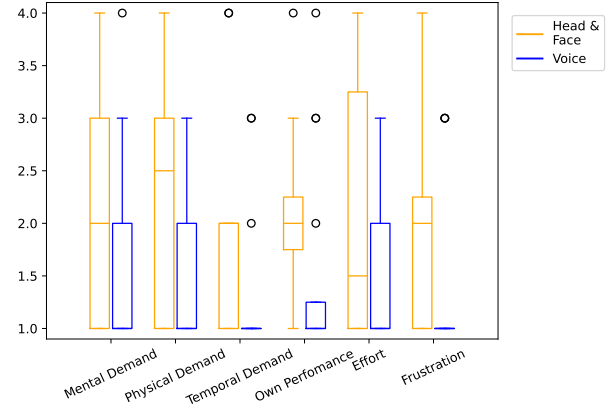
prior experience with survey input via voice compared to head movements or facial expressions. Despite these differences in user familiarity, dependent t-tests for paired samples between modalities used for the same scale granularity reveal no significant differences for any measure ($p > .05$). This indicates that our data does not show any discernible disparity in usability variables between the various modalities. This suggests that despite the differences in conveying (dis-)agreement in everyday life between head movements and facial expressions and the widespread availability of voice interfaces, participants perceived both input mechanisms for both three-point and five-point scales to be comparable in terms of usability.

NASA-TLX Scores. Figures 4a and 4b show NASA-TLX scores for 3- and 5-point scale input mechanisms respectively. While the task load is comparable between head and face input overall, the t-test for paired samples reveals a significant difference for only one measure *Mental Demand* ($t(15) = 3.22, p = .005$). In line with the findings from our initial interviews, participants therefore seemed to find it less mentally demanding, to answer Likert scale questions using head movements rather than facial expressions. The higher scores for *Physical Demand* for head input may be attributed to the reliability of detection API rather than inherent preference, as one participant noted in the final questionnaire: ‘*Head gestures mechanism can be improved a bit, we had to move heads too much to register the answer, which is too much for anyone.*’

In the case of 5-point scales, it’s understandable that voice input was perceived as less demanding compared to the combined use of head and facial expressions. Paired dependent t-tests show significant differences in mean scores for *Mental* ($t(15) = -2.30, p = .036$), *Physical Demand* ($t(15) = -3.53, p = .003$), and *Accuracy* ($t(15) = 2.2, p = .044$). Feedback regarding head movements suggests that algorithmic issues may have contributed to this discrepancy, highlighting the need for further investigation with enhanced head movement detection methods. However, for all other variables, we cannot draw conclusions regarding potential differences in perceived loads. Conducting the experiment with a larger sample size may provide more definitive insights.



(a) Nasa TLX scores for 3-point scale input mechanisms: Facial Expressions and Head Gestures



(b) Nasa TLX scores for 5-point scale input mechanisms: Combined Face and Head input, and voice

Figure 4: Nasa TLX score across modalities. Lower scores indicate lower loads and hence, better usability. Whiskers extend to the outermost points within the first to third quartile, reflecting the range of scores observed within each modality.

Input attempts. During the study, we recorded the number of overall attempts to answer each question. Mean attempts to input are reported in table 1. This measure shows that participants took more attempts during training than for any of the main sessions. Participants were given more attempts to input during training, but were able to move on from training whenever they felt comfortable with the input mechanism. The mean scores for attempts during training staying below two for all modalities shows that participants judged their own ability to navigate all of the input options to be high quickly, which is plausible given the high learnability mean scores presented in figures 3a and 3b. Interestingly, while input attempts initially decrease after training, they then increase again for the second session for facial expressions. In contrast, the mean number of input attempts continuously lowers for all other modalities as participants gather more experience, indicating they keep learning to use each mechanism.

Closing Questionnaire. After participants had used all four input mechanisms, we gathered overall preferences through a closing questionnaire. When asked which of the four input mechanisms they favor in general, 11 participants chose voice, while the remaining 5 opted for the combined face and head approach. Changing the setting to a meeting with colleagues or friends, most participants

(7) indicated they would opt for combined head and face input, as opposed to voice (4), head (3), or face (2). This preference is even more pronounced for noisy environments, in which 10 participants would opt for face+head as non-verbal input over the voice (4) or face (2) alternatives. In addition, most participants (8) chose the combination of head movements and facial expressions as the most intuitive option over voice (4), or head or face in isolation (2 each), further highlighting the potential of this combined input mechanic.

5 DISCUSSION AND FUTURE WORK

In this work, we introduced a novel hands-free method for inputting Likert scale responses by leveraging bi-modal head movements and facial expressions. Participants demonstrated successful submission of responses for 3-point Likert scale questions using either facial expressions or head movements. Both modalities received high usability ratings and were deemed low in cognitive demand, indicating their acceptance as a viable digital survey input method. In the 5-point scale scenario, the bi-modal approach exhibited comparable usability and cognitive load to the widely used voice modality. These findings underscore the considerable potential of the proposed approach for enhancing digital survey interactions. While our assessment focused on 3- and 5-point Likert scales, commonly used in digital surveys, there is potential for extension to higher granularity, such as 7-point scales and beyond. This expansion could be facilitated by synchronizing face and head movements and considering additional non-verbal modalities, such as humming.

Despite the promising results, our study uncovered certain challenges and limitations that we aim to address in future work. One area of improvement involves enhancing the accuracy of detecting facial expressions and head movements. This could be achieved through a more robust approach, potentially transitioning from a rule-based head movement detection system to a machine-learning-based one, or further stabilizing results over time. We also plan to synchronize the face and head modalities for a truly multi-modal approach, ensuring a more comprehensive understanding of users'

	Face		Head		Head+Face		Voice	
	μ	σ	μ	σ	μ	σ	μ	σ
Training	1.79	1.26	1.41	0.74	1.62	0.95	1.45	0.78
Session 1	1.24	0.56	1.24	0.58	1.41	0.59	1.23	0.53
Session 2	1.46	0.76	1.15	0.42	1.36	0.64	1.2	0.49

Table 1: Mean (μ) number of attempts to answer each question for each session and input method with standard deviation (σ). Maximum number of attempts was 3 during the main sessions and 7 during training.

responses. Additionally, the performance of modalities requires further assessment in a quantitative setting, incorporating precise measurements of time and errors.

Looking forward, we aim to expand our study settings to include a larger and more diverse participant pool in real-world environments, such as public displays. This expansion will provide valuable insights into the adaptability and effectiveness of the proposed method in various contexts. Furthermore, our research road-map includes the exploration of multi-person feedback scenarios. Leveraging head movements and facial expressions, we aim to enable users to provide feedback in social scenarios like group meetings and conferences. The envisioned application involves utilizing wall-mounted cameras to detect gestures, opening up new possibilities for quick surveys and interactive engagements in collaborative settings. This ambitious vision aligns with our commitment to advancing the capabilities and applications of bi-modal interactions in digital survey methodologies.

Moreover, expanding our research into the realms of Virtual Reality (VR) and Extended Reality (XR) offers promising prospects for continued exploration and ingenuity. By seamlessly integrating hands-free interaction method into VR/XR environments, we have the potential to craft immersive survey that transcend the limitations of conventional interfaces. This endeavor will open up fresh pathways for engagement, interaction, and data acquisition within the virtual domain, propelling survey research methodology.

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